Gcode, A Tool for the Computational Study of Complex Catalytic Pathways:

dft-GKA Investigation of Temperature-Modulated Oxide-Phase Contribution in CO/H_2 PROX over MnO_2–CeO_2 Catalysts

Francesco Ferrante,[†] Francesco Arena,[‡] Pia Domina,[†] Laura Gueci,[†] and Dario Duca*,[†]

[†]*Dipartimento di Fisica e Chimica "Emilio Segrè", Università degli Studi di Palermo, Viale delle Scienze Ed. 17, I-90128 Palermo, Italy*

[‡]*Dipartimento di Ingegneria, Università degli Studi di Messina, Contrada di Dio, I-98166 Messina, Italy*

E-mail: dario.duca@unipa.it

Gcode: How to use it

Gcode has been already employed for analyzing simpler models of the PROX catalytic reaction.[1] The example provided here, which corresponds to the version-2 commit, is currently addressed to the referees of the article indicated in the header after the colon. Gcode and its fitting extension module are described in pseudocode language in the Supporting Information of the same article. In details, the example illustrates how Gcode can be used to extract information on the PROX catalytic system at T = 473 K and a total pressure of 100 kPa, with the percentages of CO, O_2, H_2, and N_2 being 1, 1, 98, and 0, respectively.

[1]Bertini, M.; Ferrante, F.; Gueci, L.; Prestianni, A.; Duca, D.; Arena, F.; Murzin, D. Yu.. Alternative Algebraic Perspectives on CO/H_2 PROX over MnO_2 Composite Catalysts. *J. Chem. Inf. Model.* **2025**, *65*, 4952–4967.

Clearly, it could also be applied to the study of other catalytic conditions or reactions, simply modifying the `input` related to the energetic information and/or the graph representation of the involved catalytic cycles. **GCODE** for modularity and simplicity, is here shaped by two environments named WATER and FIRE. In the first, given a starting graph, representing the catalytic mechanism, the spanning tree, Σ-T, and the cycle branched component, X-T, LISP lists needed for the analysis are individuated. In the second, this analysis is performed extracting the catalytic information.

Tables and figures included in the main text of the article will be designated throughout this short note by Arabic numerals, whereas those provided in the Supporting Information will be designated by numerals preceded by an uppercase "S".

The WATER environment is formed by four component codes: **W0**, **W1**, **W2**, **W3**. The former identifies the possible branches, X-Ts, that are connected to the different cycles present in the graph. **W0**, which also allows one to determine the arms of branched cycles that originate and terminate at the same node, for each cycle present in the polycyclic catalytic mechanism graph, actually, identifies the oriented branch sets consisting exclusively of edges directed toward the cycles to which they were originally connected. These X-T branches, when isolated, have to be reconnected to the inlet nodes at the two cut points, in this way yielding the corresponding branched cycles, X-T + C(i). The associated edge sequences, grouped by reference cycle C(i), are stored in n files C_i ($i = 1, \ldots, n$), where n is the number of physically relevant cycles in the polycyclic system; $n = 7$ in the present example. Table S1 collects all the branched cycles grouped according to the corresponding reference cycles of the title reaction schematically represented in Figure 1 while examples of branched cycles are given in Figure 5. It should be noted that, irrespective of their connection points, the branches originate from cycles convoluted with the reference cycle. It should be also emphasized that the procedures involving **W0** are not fully automated. For example, in the current version of **W0**, its `main` module lists the arcs from which the branches required to construct all branched cycles — as described in the Supporting Information — can be

extracted. The arcs must be explored one at a time, following the annotations provided in the same `main` module. The remaining codes are applied, in the order: **W1**, **W2** and **W3**. In this work the result of this procedure is summarized by Tables S2–S13. In fact, **W1**, **W2** and **W3** construct the spanning trees Σ-T_i ordered by the terminal node i (see Figure 4). The resulting edge sequences are stored in L files `W2_i` ($i = 1, \ldots, L$), with `W2_0` collecting the complete set; L = 12 in the present example. **W1** is employed to extract all the graphs that could potentially give rise to the Σ-Ts (in this case, 1365 graphs). To get this result, **W1** receives as `input` the information of the file `graphD_in` summing up the list of edges representing the steps of the whole catalytic mechanism. Whereas, to reach the final information on the 378 Σ-Ts, **W2** operates on `sink_out-1`, the file `output` of **W1** containing the tuple representation of the 1365 graphs including the ones containing cycles as well as reversible edges, and **W3** on the one, `sink_out_0`, of **W2**, containing the filtered graphs not containing internal cycles but still including reversible edges. Note that by **W1** an arbitrary number of edges can be removed from the `input` graph represented in `graphD_in`. **W1** is executed with k specified as a command-line argument, where k is the number of edges to be removed, corresponding to the number of cycles present in the polycyclic starting graph structure. If unspecified, k = 2 is used as hardcoded default. **W3** writes the spanning trees to `spanning_trees_by_sink`, grouped by convergence node. It also reports on the monitor the number of spanning trees for each convergence node as well as the number of processed graphs and spanning trees identified; the former can be compared with the determinant values of the reduced Laplacian matrices obtained for the corresponding sites, in this way monitoring the correctness of the `output`. The results obtained in the WATER environment clearly do not depend on the temperature.

The FIRE environment is formed by three components, *i.e.* **F1**, **F2** and **F3**. The file `T473`, in the present example, is the `input` of **F1** when the system is analyzed at 473 K. `T473` contains the values of $\Delta G^{\ddagger}$ of the steps individuated following the node notation reported in Figure 1. The $\Delta G^{\ddagger}$ values of steps involving either adsorption or oxygen diffusion are

corrected using Equation 2. **F1**, which is executed specifying the temperature as a command-line argument, reads information from the file `T473` and loads the files `W2_0`, `W2_1` – `W2_12` and `C1` – `C7`. If the temperature is not explicitly specified, the code defaults to loading the file `T298`. `W2_0` contains the graphs of all the Σ-T lists, expressed as tuples of edges, characterizing the overall mechanism. The Σ-Ts are also separately grouped, as shown in Table S2, in the files `W2_1` – `W2_12` while `C1` – `C7` contain the same information, although referring to the different branched cycles of Table S1. These files are all processed, and the result is collected in the corresponding `END_X` files with $X = 0 - 12$, C1 – C7, which gather the related overall weights of the Σ-Ts and branched cycle, X-T + C(i), graphs expressed in terms of Total Energy Content (TEC/kJmol^{-1}) values. Each TEC $= \sum_k \Delta G_k^{\ddagger}$ is clearly related to each considered subgraph (either a spanning tree or a branched cycle), with k representing any edge/step component of the latter. **F2** is a bridge code that processes the information in the `END_X` files and returns the corresponding files `EXIT_X`, with X having the same neaning as before. These files collect in a compact form (*i.e.* without redundant information) the tuples of the TEC values for each group of Σ-Ts and branched graphs. **F3** reads the files `EXIT_X` and processes them, generating the complementary `RESULT_X` files, which report the cumulative probabilities, corresponding to the product of the exponential kernels that characterize the Σ-Ts and branched graphs, containing X-Ts. This information, through Equations 3 and 4 enables the calculation of the TOF values and surface populations and from the TOF values of the different species produced, as discussed in the article, the selectivity for given temperatures and pressures.

`END`, `EXIT` and `RESULTS` files, if present, can be safely erased and regenerated while the component codes can be linked through a batch file. Note that also **F3** is executed specifying the temperature as an integer or real command-line value. Finally, it has to be recalled that in order to evaluate the effects of the pressure of the gas phase species, useful for example in determining the reaction orders, the values of $\Delta G^{\ddagger}$ for the steps involving adsorption processes in the file `T`, in this case `T473`, must be corrected adjusting the term F of Equation 2.